

Skill Arena - Phase 5 Document 1

System Architecture, Backend Services & Microservices Design

Document Purpose

Define the high-level technical architecture used to build and operate the Skill Arena platform.

Business Objective

Provide a scalable, maintainable and fault-tolerant foundation capable of supporting global growth.

Architecture Philosophy

Service-oriented architecture with clear separation of responsibilities and independently scalable components.

Core Architecture

Client Applications, API Gateway, Microservices Layer, Event Bus, Databases, Cache Layer and Infrastructure Services.

Frontend Layer

Web Application, Mobile Applications, Admin Portal and Internal Operations Portal.

API Gateway

Single secure entry point for clients, authentication, rate limiting and request routing.

Microservices Strategy

Independent services responsible for gameplay, wallets, tournaments, profiles, notifications and security.

User Service

Manages accounts, profiles, authentication references and user lifecycle events.

Gameplay Service

Manages challenges, matches, gameplay validation and challenge state.

Matchmaking Service

Handles queueing, player matching and challenge allocation.

Tournament Service

Manages tournaments, championships, brackets and event registrations.

Wallet Service

Manages balances, transactions, rewards and wallet operations.

Treasury Service

Handles reserves, reconciliation, solvency and treasury operations.

Replay Service

Stores replay metadata, verification records and replay retrieval.

Notification Service

Delivers email, in-app and future push notifications.

Security Service

Handles risk scoring, device intelligence, fraud signals and enforcement workflows.

Compliance Service

Supports KYC, AML reviews and compliance operations.

Event Bus Architecture

Asynchronous communication between services using event-driven design.

Database Strategy

Dedicated databases per service domain with controlled data ownership.

Caching Layer

Accelerates frequently accessed data, rankings, profiles and session information.

Observability Framework

Centralized logging, metrics, tracing and monitoring across all services.

Fault Tolerance

Graceful degradation, retries, circuit breakers and service isolation.

Scalability Requirements

Services scale independently based on workload demand.

Security Requirements

Service-to-service authentication, encryption and least-privilege access controls.

Deployment Requirements

Support cloud-native deployments, rolling updates and automated scaling.

Acceptance Criteria

Platform architecture supports growth, resilience, maintainability and operational visibility.